

How Far Off the Recorded Path Can a Gaussian-Splat Driving Simulator Be Trusted?

CS 566 Project Proposal (one document per team) · Kyle Sha (ID 9085525476) · Allen Xia (ID 9085517234) · September 29, 2026

Main papers: [1] **Difix3D+** (CVPR 2025 Oral), *re-implemented* · [2] **FaithFusion** (CVPR 2026), *compared* · [3] **HUGSIM** (TPAMI 2026), *simulator for evaluation*

1. Problem Statement

Autonomous-driving software is increasingly evaluated in simulators reconstructed from real driving logs. The recorded video is converted into a **3D Gaussian splat** [13], a scene representation composed of millions of small anisotropic Gaussians that can be rendered from arbitrary camera poses, so that the simulated vehicle may follow a trajectory different from the recorded one. However, **a reconstruction is reliable only near the recorded trajectory**. When the simulated vehicle changes lanes or the camera is mounted higher (e.g., on a truck), rendered images degrade, and a driving system evaluated on them may behave differently than it would in reality. This project addresses two questions: **how far from the recorded trajectory a splat-based simulator remains trustworthy, and whether current generative repair methods ("fixers") restore trustworthiness or merely improve visual realism**. We define **trust** as the agreement between the outputs of a lane detector and a vehicle detector on a rendered image and on the **real** image captured at the same pose.

2. Significance and Motivation

Simulation is a principal means of validating driving software at scale. If renders away from the recorded trajectory are unreliable, a simulator may accept unsafe behaviour or reject safe behaviour, which undermines its value for validation. Establishing the range over which such simulators can be trusted is therefore of direct practical importance. The problem is also well suited to this course, as it combines camera geometry, 3D reconstruction, image restoration, and CNN-based perception.

3. State of the Art

- **Splat-based driving simulators:** HUGSIM [3] executes complete driving policies in reconstructed scenes; the EUVS benchmark [7] shows that current methods degrade substantially under large viewpoint changes.
- **Diffusion-based fixers:** Difix3D+ [1] repairs off-trajectory renders with a single-step diffusion model and distills the repaired views back into the 3D scene, reporting an approximately $2\times$ FID improvement over prior methods. NVIDIA Fixer [4] applies the same approach to driving data, and FaithFusion [2] restricts how much a fixer may overwrite the reconstruction using a per-pixel confidence map.
- **Evaluation practice:** these methods are evaluated primarily with image-realism metrics (FID, LPIPS) or with shifted cameras for which no real image exists (PRG-Fusion [6]). A recent study [10] measures perception degradation versus viewpoint offset but does not evaluate fixers. SetDiff [11] and MV2 [9] use real off-trajectory images, but only with image-similarity metrics, without comparing fixers or quantifying hallucinated content. **To our knowledge, no prior work compares fixers against real off-trajectory images in terms of the content they invent.**

4. Proposed Approach

- **Re-implementation:** we re-implement Difix3D+ [1], comprising the fixer's training procedure and its progressive update that distills repaired views into the 3D scene. The implementation is written from the paper on the same pretrained diffusion model used by the authors, validated against NVIDIA's official code and weights, and then trained on driving data.
- **New contributions:** (a) an **evaluation against real ground truth**, using Para-Lane [8], which scans the same roads in three parallel lanes (approximately 3 m and 6–7.7 m apart), and MV2 [9], in which a car and a scooter record the same road at different camera heights; training pairs are likewise constructed from neighbouring real lanes (cf. ReCamDriving [12]); (b) **hallucination accounting**, measuring invented or erased lane markings and spurious vehicles in addition to image similarity; (c) a **lightweight fixer trained from scratch** (a 15M-parameter U-Net, approximately $60\times$ smaller than the Difix diffusion backbone); (d) a **geometry-only road-surface prior**, adapted from ViSE [15], which regularizes the road using LiDAR and cannot introduce new content; and (e) a **cross-dataset transfer study** on data from other countries and camera rigs (EUVS/MARS [7], MV2 [9]).

5. Limitations of Existing Approaches and Expected Advantages

Existing fixers are diffusion models trained to produce realistic images; they are therefore rewarded for plausible detail even where that detail is incorrect. A preliminary study (September 27–28) is consistent with this: at a 3 m lateral offset, lane-detector agreement with the real image falls from 0.98 to 0.07, and NVIDIA's released Difix improves image similarity (LPIPS 0.52 $\rightarrow$ 0.44) but not lane-marking precision (0.52 $\rightarrow$ 0.48). We test three hypotheses. **(H1)** Training a fixer on real adjacent-lane pairs makes it more faithful, not only more realistic; in the preliminary study, our re-implemented Difix raised lane agreement from

0.07 to 0.71 on two held-out roads, at the cost of more spurious vehicles (0.18 versus 0.04 per image). **(H2)** A geometry-only prior improves lane fidelity without introducing hallucinated content. **(H3)** Fixers trained on one city do not transfer to others, so trustworthiness must be established per domain.

6. Evaluation Plan and Expected Results

- **Deviation axes:** lateral offset (primary; real images at approximately 3 m and 6–7.7 m), camera height (primary; real car-scooter pairs from MV2, with a road-plane-warped reference on Para-Lane as a fallback), and heading (secondary; exact image warp, restricted to regions observed by the original camera).
- **Metrics:** image similarity to the real image (PSNR, LPIPS); lane-marking precision and recall (invented versus erased markings); lane-detector (CLRNet [14]) and vehicle-detector (YOLO11) agreement, following the metrics of ReconDreamer [5]; spurious-vehicle count; and, as an extension, a driving policy's score within HUGSIM [3].
- **Methods compared:** raw render, released Difix, NVIDIA Fixer, our re-implemented Difix, our lightweight U-Net, progressive update, FaithFusion weighting, and the road-surface prior.
- **Expected results:** *Figure 1*, trust versus lateral offset for each method; *Figure 2*, trust versus camera height; *Table 1*, hallucination statistics per method; *Table 2*, cross-dataset transfer; *Table 3*, our Difix re-implementation versus the official code; and side-by-side videos with hallucinated content annotated.

Component	In the paper	Our version
Difix fixer [1]	Diffusion fixer; official code and weights released	Re-implemented and retrained by us on driving pairs; checked against official
Progressive update [1]	Fixed images fed back into the 3D scene	Re-implemented
FaithFusion [2], NVIDIA Fixer [4]	Released code / weights	Run as baselines
Lightweight fixer	—	New, trained from scratch
Road prior [15]	Road-surface model (ViSE)	Adapted and re-implemented

7. Timeline

Dates	Kyle Sha	Allen Xia
9/29–10/12	MV2 reconstruction and camera-height evaluation; additional Para-Lane test scenes	Evaluation framework for MV2 and EUVS; per-scene variance analysis
10/13–10/29	Validation of the Difix re-implementation against the official code	Road-surface prior on all scenes; hallucination analysis; mid-term report (10/29)
10/30–11/20	Cross-dataset transfer study; progressive update on all scenes	Closed-loop evaluation in HUGSIM (extension)
11/21–12/10	Final figures and videos	Presentation (12/1–12/10) and project webpage (12/10)

Risks: the preliminary results rest on few test roads, so additional scenes will be added and per-scene variance reported; some driving policies require a legacy software stack, so closed-loop evaluation uses the LTF policy. Project webpage: ksha23.github.io/cs566-drive-sim.

References

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3. HUGSIM, TPAMI 2026 · arxiv.org/abs/2412.01718 · github.com/hy Zhou404/HUGSIM
4. NVIDIA Fixer, 2025 · github.com/nv-tlabs/Fixer
5. ReconDreamer (detector-agreement metrics), CVPR 2025 · arxiv.org/abs/2411.19548
6. PRG-Fusion, 2026 · arxiv.org/abs/2609.06948
7. EUVS benchmark, ICCV 2025 · arxiv.org/abs/2412.05256
8. Para-Lane dataset, 3DV 2025 · arxiv.org/abs/2502.15635
9. MV2 dataset, ECCV 2026 · arxiv.org/abs/2608.12442
10. From Concept to Capability (3DGS for AD), SafeComp 2026 · arxiv.org/abs/2605.01995
11. SetDiff, 2026 · arxiv.org/abs/2601.07540
12. ReCamDriving, ECCV 2026 · arxiv.org/abs/2512.03621
13. 3D Gaussian Splatting, SIGGRAPH 2023 · arxiv.org/abs/2308.04079
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